🧬 HBB Gene and E6V Mutation

A Bioinformatics Analysis Project

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1. **Introduction**

The HBB gene (Hemoglobin Subunit Beta) is responsible for encoding the beta chain of hemoglobin, the protein that carries oxygen in red blood cells. Mutations in this gene can lead to severe blood disorders, the most well-known being Sickle Cell Disease (SCD).

One of the most important mutations in the HBB gene is the E6V mutation, where the amino acid Glutamic acid (E) is replaced by Valine (V) at position 6 of the beta-globin protein. This project presents a simple bioinformatics analysis of the HBB gene and this critical mutation.

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2. **Gene Information**

Gene Name: HBB

Full Name: Hemoglobin Subunit Beta

Chromosomal Location: Chromosome 11 (11p15.4)

Organism: Homo sapiens

Protein Length: 146 amino acids

Main Function: Oxygen transport in red blood cells

The HBB gene combines with the alpha-globin gene to form functional hemoglobin (HbA).

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3. **E6V Mutation** (Sickle Cell Mutation)

Mutation Type: Missense mutation

Position: Amino acid 6

Normal Amino Acid: Glutamic acid (E)

Mutated Amino Acid: Valine (V)

Mutation Name: E6V

Genetic Change: GAG → GTG

Disease: Sickle Cell Anemia

This mutation changes the chemical properties of hemoglobin, causing red blood cells to become rigid and sickle-shaped under low oxygen conditions.

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4. **Sequence Analysis Using BLAST**

The HBB nucleotide and protein sequences were analyzed using NCBI BLAST.

BLAST Results Summary:

High similarity with human HBB reference sequence

Strong alignment with primate species

Moderate similarity with mouse beta-globin

Very low E-value, indicating a highly significant match

BLAST confirms that the analyzed sequence correctly corresponds to the HBB gene.

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5. **Protein Analysis Using UniProt**

UniProt provides detailed functional annotation for the HBB protein:

Protein Name: Hemoglobin beta chain

Molecular Function: Oxygen binding

Biological Role: Gas transport, erythrocyte function

Post-translational Modifications: Heme binding sites present

Structural Feature: Highly conserved beta-globin fold

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6. **Conserved Regions and Multiple Sequence Alignment**

Using Clustal Omega, multiple sequence alignment was performed between:

Human

Mouse

Chimpanzee

Key Observations:

High conservation in functional regions

The N-terminal region (including position 6) is highly conserved

This confirms the biological importance of the E6 position

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7. **Biological Impact of the E6V Mutation**

The E6V mutation causes:

Reduced hemoglobin solubility

Polymerization of deoxygenated hemoglobin

Deformation of red blood cells

Reduced oxygen delivery

Pain crises, anemia, and organ damage

This shows how a single nucleotide change can cause a severe systemic disease.

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8. **Conclusion**

This project demonstrates the power of bioinformatics tools in understanding genetic mutations and their biological effects. The analysis of the HBB gene and the E6V mutation highlights the relationship between DNA sequence, protein structure, and human disease.

Bioinformatics allows researchers to study genetic disorders even without access to a physical laboratory.

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9. **Tools Used**

NCBI Gene

NCBI BLAST

UniProt

Clustal Omega

VEP (Variant Effect Predictor)